

TECHNICAL VALIDATION REPORT

Redacted Public Version (TRL-6)

1. Mathematical Proof of Efficacy

Comparison of Linear Weighted Sum (WSM) vs MGOS Nonlinear Logic (Phi).

Scenario: Asymmetric Cyber Attack (Cyber=0.96, Others=0.1)

Standard Linear Model Calculation:
Score = (Sum of weighted inputs) / Total Weights
Result = 0.2720
Verdict: FALSE NEGATIVE (System ignores threat)

MGOS GOLD Phi(x) Logic:
IF critical_sensor > threshold THEN override = 1.0
Result = 1.0000
Verdict: TRUE POSITIVE (Critical Alert)

2. Empirical Verification Log

Actual logs from the TRL-6 Engine run on 30.11.2025.

[TEST CASE: CYBER_ASYMMETRIC]
Input Vector Loaded: {Cyber: 0.96, OSINT: 0.1, ...}
Processing Layer: STCL Fractal Core
> Linear Check: 0.27 (Low)
> Nonlinear Check: CRITICAL OVERRIDE TRIGGERED
Final Output: DEFCON 1

3. Safety Mechanisms (Dead Man's Switch)

[TEST CASE: SENSOR_FAILURE]
Input Vector: All Zeros (0.0)
Expected: DEFCON 5 (Peace) in linear models
MGOS Output: ERROR CODE 4 (Sensor Malfunction)
Result: PASS